

UNIVERSITY-PATTERN NUMERICAL PROBLEMS WITH SOLUTIONS

Static Inventory Problems under Risk

NUMERICAL PROBLEM 1 – DISCRETE DEMAND (EXPECTED COST METHOD)

A newspaper vendor must decide how many copies to order. Demand and probabilities are given below.

Demand (copies): 40, 50, 60
Probability: 0.2, 0.5, 0.3

Purchase cost = ₹5 per copy
Selling price = ₹8 per copy
Unsold copies have no value.

SOLUTION:

Shortage cost C_u = Selling price – Purchase cost = $8 - 5 = ₹3$
Excess cost C_o = Purchase cost = ₹5

Step 1: Construct cost table
Step 2: Compute expected cost for each order quantity
Expected Cost = $\sum (\text{Probability} \times \text{Cost})$

The order quantity with minimum expected cost is optimal.

NUMERICAL PROBLEM 2 – TOTAL COST MATRIX

A shopkeeper considers ordering 50, 100, or 150 units. Possible demand levels are 50, 100, and 150 units.

Shortage cost = ₹6 per unit
Excess cost = ₹2 per unit

SOLUTION:

Construct total cost matrix by computing excess or shortage cost for each decision-demand pair.
Choose order quantity based on minimum expected or minimax regret criterion.

NUMERICAL PROBLEM 3 – OPPORTUNITY COST MATRIX

Using the data from Problem 2, construct opportunity cost matrix and apply minimax regret rule.

SOLUTION:

Step 1: Identify minimum cost in each demand column
Step 2: Subtract minimum from each element
Step 3: Find maximum regret for each decision
Step 4: Choose decision with minimum regret

NUMERICAL PROBLEM 4 – COST OF RISK

Expected cost under risk = ■120

Cost under certainty = ■40

Find the cost of risk.

SOLUTION:

Cost of Risk = Expected Cost – Cost under Certainty

= 120 – 40 = ■80

NUMERICAL PROBLEM 5 – CONTINUOUS DEMAND (CRITICAL RATIO)

A fashion retailer sells a seasonal jacket.

Purchase cost = ■600

Selling price = ■1,000

Salvage value = ■400

SOLUTION:

C_u = Selling price – Purchase cost = 1000 – 600 = ■400

C_o = Purchase cost – Salvage value = 600 – 400 = ■200

Critical Ratio:

$P(D \leq Q) = C_u / (C_u + C_o)$

= 400 / (400 + 200)

= 0.667

The retailer should order quantity corresponding to 66.7% cumulative demand.